

Simon Degala Campos

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SUMMARY

Senior Software Engineer | Backend & Python Expert with 6+ years of specialization in designing and building scalable, high-performance backend systems. Skilled AI/ML Engineer with a strong track record in developing robust AI agents and intelligent solutions. Proficient in creating efficient, scalable architectures that drive innovation and support complex AI workflows. Adept at leveraging cutting-edge technologies to deliver impactful, reliable backend services for demanding applications.

SKILLS

AL & ML

PyTorch, TensorFlow, Keras, JAX, Hugging Face, LangChain, RAG, LlamaIndex, Numpy, Pandas, Polars, Scikit-learn, Jupyter, OpenCV, XGBoost, MLflow, Streamlit

Frontend

React, Next.js (App Router, Server Components, Turbopack), JavaScript (ES2025+), TypeScript 5.6+, HTML/CSS (CSS Nesting, Container Queries), Tailwind CSS 4, Zustand, Vite 6, Yarn, Figma

Backend

Python, Django, FastAPI, Flask, Node.js, Express.js, Go, Rust, RESTful API, GraphQL, WebSockets, Microservices, MySQL, PostgreSQL, MongoDB, Redis, Supabase

Mobile

Flutter, Dart, React Native, Kotlin, Swift, Android Studio, Xcode, Firebase

DevOps

AWS, Google Cloud, Kubernetes, Github Actions, Docker, Terraform, Kafka, Grafana, RabbitMQ, Git, Github, Gitlab

Trading

Timeseries Forecasting (LSTM, TimeXer, TimeMixer, TimeMixer++, TabPFN-v2, TIME-MOE), Sentiment Analysis, Pattern Detection, Algorithm Trading, Backtesting, Optimization, Cryptocurrency, Forex Trading, Stock Market, Trading Automation (TradingView, IBKR, MetaTrader 4, MetaTrader 5, NinjaTrader 8, cTrader, Binance, Bybit, Bitget, Blofin, MEXC, TradeStation)

Other

Cursor, Postman, Selenium, Cypress, OpenAI API,

EXPERIENCE



Founder & CTO

PXI Labs LLC

📅 02/2026 - Present 📍 Remote

PXI Studio is a social and creative workspace platform: users get a ranked media feed, events and ticketing, albums with threaded discussion, chat, scrapbooks, a vault, notifications, and vendor/event-staff flows - delivered as a native mobile app, a marketing and dashboard web app, and a dedicated API tier.

- Led engineering end to end as Founder / CTO / Lead Software Engineer: owned system design across three codebases (Expo/React Native mobile, Next.js web, Node/Express API), data model split (relational + document), auth strategy, media pipeline, payments, and deployment story (e.g. API on EC2 with Docker/CI per repo docs; web on Netlify with Edge middleware).
- Designed and shipped a TypeScript/Express backend with REST APIs for auth, feed, social graph, events, albums, chat, media, scrapbooks, vault, tickets, vendors, safety, and webhooks; PostgreSQL (Prisma) + MongoDB (Mongoose) for the right data on the right store; Socket.io for real-time behavior where the product needs it.
- Implemented security-sensitive auth: PASETO v4 token issuance and verification, integration with Next.js Edge middleware and public-key verification for dashboard RBAC without extra round-trips; social login (Google/Apple) and session patterns wired through mobile and web clients.
- Built feed and engagement mechanics including Wilson score-based ranking, reactions, and notification flows so discovery and quality signals are first-class in the API and clients.

- Owned media and storage architecture using AWS S3 (upload SDKs, presigned flows) so images/video and related assets scale; paired with mobile capture/edit UX and backend media routes.
- Integrated monetization and operations: Stripe (including Connect-style flows and a unified webhook surface for platform + connected accounts), plus transactional email (Resend) and SMS/voice (Twilio) where the product requires it.
- Shipped the mobile app on Expo / React Native (Expo Router, TypeScript, NativeWind): tabs for core surfaces (e.g. wall, studio, camera, passport), secure token storage, socket.io-client, Stripe React Native, camera stack (react-native-vision-camera), and advanced imaging (React Native Skia shaders/LUT-style processing for the “studio” camera pipeline).
- Delivered the web property on Next.js (App Router) with public marketing/event pages and a middleware-protected dashboard (passport, vendor upgrade, account); Ant Design, charts (Recharts), and Stripe.js for web payments; aligned with backend “soft gate” vs verified-user rules documented in the API guide.
- Established how the team ships: API and DB documentation, dev workflow (branching/PR policy), and CI/CD notes so contributors can run, migrate, and deploy consistently—reducing bus factor as the org grows.



AI Full Stack Developer

Upwork

📅 06/2024 - 02/2026 📍 Remote

Upwork is a leading global freelancing platform headquartered in Palo Alto, California, connecting businesses with independent professionals and agencies across various categories including software development, design, marketing, and more.

- Worked as a freelance full-stack developer on Upwork, completing 10+ client projects ranging from AI-powered web and mobile apps to custom trading tools and automation systems
- Built end-to-end web applications with React/Next.js on the frontend and Node.js or Python backends, exposing REST/GraphQL APIs and integrating real-time features where needed
- Developed AI-driven features such as chatbots, recommendation flows, and predictive dashboards by wiring custom or third-party models into backend services and frontend UX
- Created cross-platform mobile apps using Flutter and/or React Native, integrating backend APIs and, in some cases, lightweight on-device inference or remote model calls for AI functionality
- Designed and implemented several algorithmic trading systems for crypto/forex, including data ingestion, strategy backtesting, live execution, and risk controls, using Python and Go on the backend
- Worked with common ML libraries and services (TensorFlow/PyTorch, scikit-learn, or hosted APIs) to build or integrate models for forecasting, classification, and basic NLP tasks, then deployed them behind secure APIs
- Containerized backend services with Docker and used CI/CD workflows to deploy to AWS/GCP or other hosting platforms, helping clients ship updates more frequently with fewer manual steps
- Collaborated closely with non-technical clients to clarify requirements, propose technical approaches, estimate work, and provide regular progress updates, ensuring projects were delivered on time and within scope



Senior Backend Developer

Plentina VN

📅 04/2022 - 05/2024 📍 Paranaque, Philippines

Plentina is a fintech startup providing Buy Now, Pay Later (BNPL) credit solutions in emerging markets, primarily in the Philippines. The company helps consumers build credit history and make cashless payments through a mobile app and partnerships with merchants

- Led backend development for credit and loan management services, designing and maintaining core services that handled user onboarding, credit assessment, repayment schedules, and transaction processing
- Worked with modern backend frameworks and cloud services (Node.js/TypeScript, PHP/Laravel, REST APIs, Kafka, Kubernetes, Docker) to build scalable and resilient systems that supported growing user and transaction volume
- Designed and maintained high-throughput RESTful APIs used by the mobile app, internal tools, and partner integrations, with a focus on clear contracts, versioning, and backward compatibility
- Collaborated closely with data and product teams on risk and credit features, integrating scoring logic, limits, and eligibility checks directly into the backend flows to reduce manual review and improve approval accuracy
- Implemented and optimized backend components that consumed machine learning outputs (risk scores, predictions, segment tags) for credit decisioning and personalized limits, making sure inference calls were reliable, cached appropriately, and observable in production
- Helped design secure data flows for sensitive customer and transactional data, including encryption at rest and in transit, strict access controls, and proper logging/auditing aligned with internal policies and local financial regulations
- Tuned database schemas and queries (MySQL/PostgreSQL) to handle high- volume, low- latency BNPL transactions, including adding the right indexes, breaking down heavy queries, and separating read/write workloads where needed
- Built and maintained integrations with third-party payment gateways, e-wallets, and merchant systems, handling edge cases such as timeouts, partial failures, idempotency, and reconciliation of transaction records
- Took ownership of incident handling for backend services, including setting up monitoring and alerts, performing root-cause analysis on production issues, and implementing permanent fixes instead of short-term workarounds
- Worked with product managers and mobile engineers to break down feature requirements, estimate work, and plan releases; participated in code reviews and helped maintain consistent coding standards and documentation across the backend team
- Contributed to CI/CD pipelines and deployment practices such as automated tests, staging environments, canary or phased rollouts, which helped reduce deployment risk and improve release frequency
- Assisted in evolving the system architecture over time, gradually refactoring and extracting services from a more monolithic structure into a cleaner, more modular backend that could support new products and markets



Junior Backend Developer

Converge ICT

📅 10/2019 - 04/2022 📍 Angeles, Philippines

Converge ICT Solutions Inc. is a premier fiber optic internet and digital services provider in the Philippines, known for its extensive fiber network and commitment to reliable, high-speed internet connectivity for homes and businesses

- Developed and maintained backend features for internet service provisioning and billing platforms using Node.js, PHP, and related frameworks, helping support day-to-day operations and improve the overall customer experience
- Worked on MySQL and PostgreSQL databases by designing and refining schemas, writing optimized queries, and handling large sets of customer and service data to reduce query times and improve application responsiveness.
- Built and integrated secure RESTful APIs with third-party monitoring, provisioning, and billing systems, focusing on reliable data exchange, proper error handling, and consistent API documentation
- Investigated and resolved backend performance issues, such as slow endpoints and inefficient queries, and helped address security concerns by applying patches, tightening access controls, and following internal security guidelines
- Collaborated with senior engineers to design and implement modular features, participated in sprint planning and stand-ups, and followed agile practices to deliver updates in small, manageable increments
- Contributed to the CI/CD pipeline by writing and maintaining build scripts, adding basic automated tests, and assisting with deployments to staging and production environments
- Performed code reviews for peer changes, followed established coding standards, and helped refactor legacy code to make it easier to maintain and extend
- Assisted in upgrading and migrating parts of the legacy backend stack to more scalable and maintainable solutions, helping the team support new services, higher traffic, and additional business requirements

EDUCATION

Bachelor's Degree in Computer Science

University of Rizal System (URS)

📅 2016 - 2019 📍 Tanay, Philippines

- Completed coursework in software development, data structures, algorithms, databases, and programming languages
- Participated in projects involving web development, mobile apps, and software design
- Gained practical experience in coding, debugging, and collaborative development environments

PROJECTS

PXI Studio - Social & Creative Workspace Platform (Full-Stack: Mobile, Web & API)

🔗 <https://www.pxispace.com/>

A multi-client platform for events, media, and community: ranked social feed, albums and threaded discussion, real-time chat and notifications, ticketing and vendor/staff flows, scrapbooks, vault, and account verification—delivered as a native mobile app, a Next.js marketing and dashboard site, and a dedicated Node API.

Technical Features

- Led end-to-end architecture across three repositories (Expo/React Native, Next.js, Express/TypeScript) with a clear split between product surfaces and a shared REST + real-time backend.
- Built a TypeScript Express API with PostgreSQL (Prisma) and MongoDB (Mongoose) so relational identity/settings and high-churn social/document data each sit on an appropriate store; documented APIs, schema, and deployment for the team
- Implemented PASETO v4 token auth and wired Next.js Edge middleware to verify tokens and enforce dashboard RBAC without extra round-trips; integrated Google/Apple sign-in and mobile secure token storage
- Shipped a Wilson score-ranked media feed with reactions, notifications, and social graph primitives (follows, friend requests, safety/reporting patterns as implemented in routes)
- Added Socket.io for real-time behavior alongside REST, plus AWS S3-backed media upload and delivery patterns for images and user-generated content
- Integrated Stripe (including Connect-oriented flows and a unified webhook surface), plus Resend and Twilio for operational messaging where the product requires it
- Developed the Expo/React Native app with Expo Router, TypeScript, and NativeWind: camera capture, advanced on-device imaging (React Native Skia / vision-camera stack), albums and thread UI, checkout hooks, and socket client integration
- Delivered the Next.js (App Router) web app with public pages and middleware-protected dashboards (passport, vendor upgrade, account), Ant Design UI, charts (Recharts), and Stripe.js for web payments
- Established CI/CD (e.g. API on EC2 with Docker/GitHub Actions per project docs; web on Netlify), environment conventions, and contributor workflow so releases stay repeatable

Technology Stack

TypeScript, Node.js, Express, Prisma, PostgreSQL, MongoDB, Mongoose, PASETO, Socket.io, AWS S3, Stripe, Next.js, React, React Native, Expo, NativeWind, Ant Design, Docker, GitHub Actions, Netlify (Edge middleware)

FxUtopia - Forex & Crypto Trading Hub API (High-Performance Python Backend)

🔗 <https://www.fxutopia.com/>

A comprehensive FastAPI backend for a Forex & Crypto trading community platform featuring authentication, community forums, educational content, marketplace, reviews, signals, and economic calendar.

Technical Features

- Architected a modular FastAPI backend for a Forex & Crypto trading platform using Clean Architecture with repository/service layers to keep business logic isolated and easy to test
- Implemented secure auth flows with JWT access/refresh tokens, Google/GitHub OAuth, email verification, password reset, 2FA, and role-based access control
- Built community features including posts, threaded comments, reactions, user badges, and WebSocket chat, with pagination, caching, and full-text search for scalable read patterns
- Delivered education modules for courses, articles, and videos with progress tracking, completion certificates, and a searchable knowledge base
- Developed a marketplace for digital trading tools (EAs, bots, indicators) covering product listing APIs, secure file upload/delivery, and integrated ratings and reviews, designed to plug into a payment gateway
- Implemented broker/platform review flows with verified reviews, detailed rating dimensions, and “helpful” votes to surface quality content
- Exposed trading signal and provider APIs with subscription management, performance stats, leaderboards, and WebSocket endpoints for near real-time distribution
- Added an economic calendar API for macro events with impact levels, country/currency filters, and live update support
- Inter-Service Communication: gRPC protocol implementation
- Used Redis for caching and Celery workers/beat for background and scheduled tasks (emails, analytics, file processing) to keep the main API responsive
- Designed the relational data model in PostgreSQL and managed migrations with SQLAlchemy + Alembic, including indexes and constraints aligned with query access patterns
- Introduced structured logging, metrics, centralized error handling, and rate limiting plus security middleware (CORS configuration, strict validation) for production readiness
- Containerized services with Docker and Docker Compose, wiring up API, PostgreSQL, Redis, and Celery into a reproducible local and deployment setup

Technology Stack

Python, Django, FastAPI, PostgreSQL, Redis, SQLAlchemy, Alembic, Celery, JWT, Pydantic, Docker

Jobs3 - Next-Gen Job Marketplace with AI and Blockchain (Large-scale Node.js Backend)

A comprehensive backend system for Jobs3, a modern job marketplace platform similar to Upwork. This Node.js/Express backend provides a full-featured API for connecting freelancers, employees, employers, and clients with intelligent job matching, real-time communication, and AI-powered search capabilities.

Technical Features

- Supports four distinct roles (Freelancers, Employees, Employers, Clients) with role-specific dashboards, permissions, and workflows, enabling flexible marketplace dynamics similar to Upwork's hybrid model
- Full CRUD operations for diverse job types (fixed-price, hourly) across client gigs, employer postings, freelancer services, and employee opportunities, with advanced filtering by skills, location, and budget
- Socket.io-powered bidirectional chat with online status, message history, and typing indicators, ensuring seamless client-freelancer communication comparable to modern freelance platforms
- OpenAI GPT-4 integration with LangChain for semantic matching, generating intelligent recommendations based on skills, experience, and job requirements beyond keyword search
- Pinecone vector database enables semantic search across job descriptions, profiles, and portfolios, delivering context-aware results that improve match accuracy by understanding intent
- End-to-end workflow from proposal submission (with bids, timelines, rates) through contract creation, extensions, milestones, and dispute handling, streamlining the freelance lifecycle
- Freelancers/employees showcase multimedia portfolios with visit analytics, paired with a bidirectional rating system that builds platform trust through verified feedback
- Node-cron scheduled tasks maintain real-time data consistency across MongoDB and Pinecone, automatically indexing new jobs for AI search without manual intervention
- AWS S3 integration with Sharp image optimization and Multer handling supports unlimited portfolio uploads, resumes, and contract documents with CDN delivery
- Nodemailer email campaigns plus Telegram bot integration deliver instant alerts for proposals, contract updates, job matches, and payment milestones across user preferences

Technology Stack

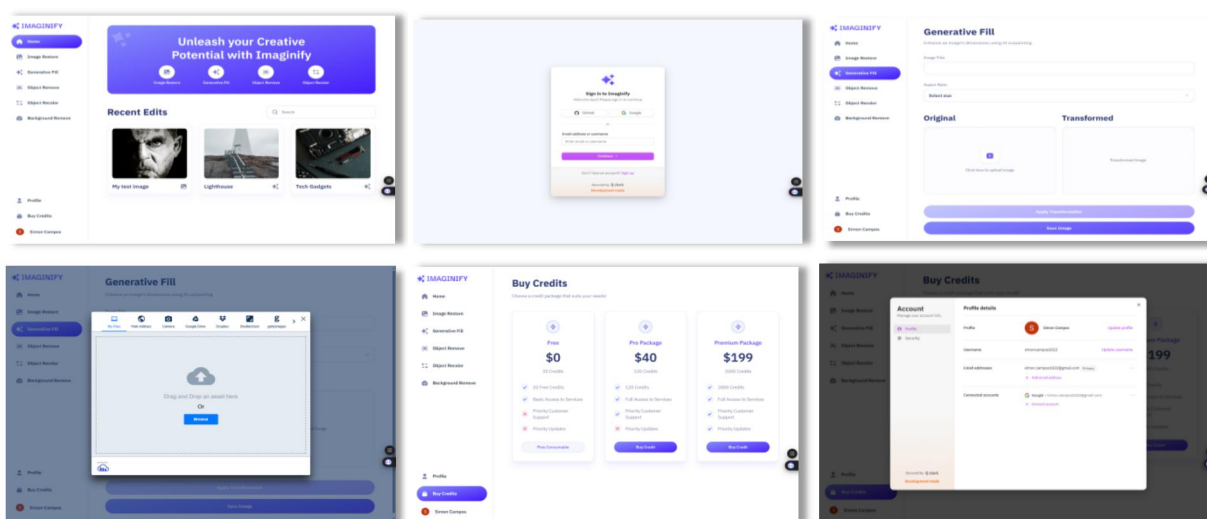
Node.js, Express.js, MongoDB, Socket.io, JWT, OpenAI API, Pinecone, LangChain, S3, GridFS, Sharp, Multer, Bcryptjs, Node-Cron, Compression, Morgan, Nodemailer

Imaginify - Image Transformation Platform (AI-Powered SaaS App)

<https://ai-sass-app-pi.vercel.app/>

Imaginify is a cutting-edge AI-powered SaaS platform that transforms images using advanced machine learning algorithms. Built with modern web technologies, it provides users with powerful image editing capabilities through an intuitive and beautiful interface. The platform leverages state-of-the-art AI models to deliver professional-grade image transformations that were previously only available to experts with expensive software.

- Imaginify is an AI-powered SaaS platform that delivers professional-grade image transformations using advanced machine learning models like CNNs, GANs, and Stable Diffusion.
- It offers real-time, high-quality image restoration, background removal, object removal, content-aware expansion, and style transfer with support for up to 8K resolution and batch processing
- The platform features a flexible subscription system with credit management and secure Stripe payment integration for seamless user experience
- Imaginify provides robust user management including multi-factor authentication, social logins, API key control, and AI-powered image organization with smart collections and semantic search
- Built with modern technologies such as Next.js, React, TypeScript, Tailwind CSS, and MongoDB, it ensures a responsive, accessible, and intuitive UI with progressive web app capabilities
- A complete backend architecture supports fast GPU-accelerated AI inference, Redis caching, serverless edge functions, and comprehensive error monitoring for reliable performance at scale

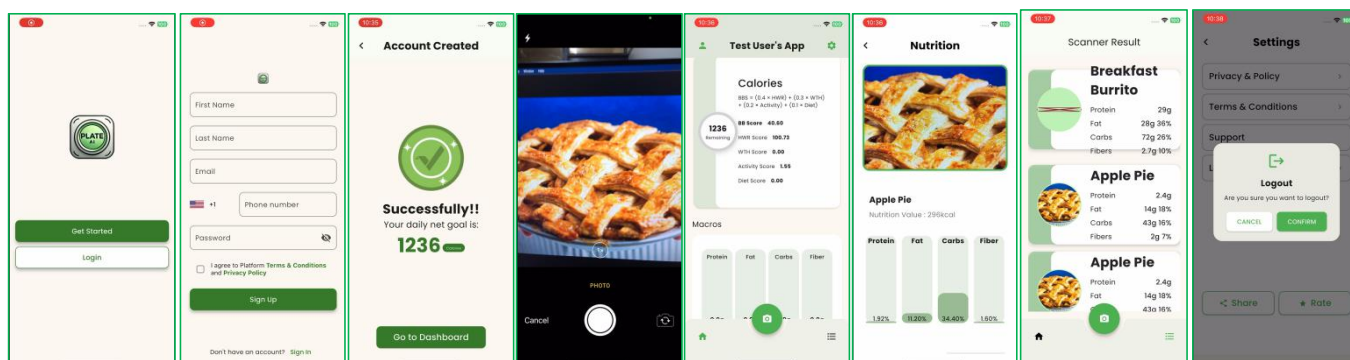


Plante AI - Smart Meal Analysis (AI-Powered Mobile App)

<https://drive.google.com/drive/folders/1T0s3laE9Z6iulVpn4Z99IYFK7xCbErO?usp=sharing>

Plante AI is an innovative Flutter application that revolutionizes how we understand and improve our eating habits through advanced AI-powered meal analysis. Simply take a photo of your meal, and our intelligent system will provide comprehensive insights, health recommendations, and safety alerts.

- Plante AI is a Flutter app that uses AI to analyze meals from photos, providing detailed nutritional information and calorie estimates.
- It offers personalized health and dietary recommendations based on user goals and nutritional gaps. The app detects allergens, unsafe food combinations, and contamination risks to ensure meal safety
- The app detects allergens, unsafe food combinations, and contamination risks to ensure meal safety
- Users can track their meal history, monitor dietary trends, and receive an overall health score for their eating habits
- The app supports offline use, multiple languages, and runs seamlessly on iOS, Android, and Web platforms



Botara 3.0 - Advanced Futures Trading Strategy

<https://github.com/simoncamos1022/profitable-futures-trading-strategy-crypto-forex-stock>

A sophisticated algorithmic trading system designed for cryptocurrency and forex futures markets. This strategy implements a multi-indicator strategy with advanced risk management, position tracking, and real-time monitoring capabilities.

- Designed and implemented a fully automated multi-asset futures trading bot (crypto, forex, stocks) using Python 3
- Built a multi-indicator strategy combining Fisher Transform, RSI, Volume Oscillator, and ATR with dynamic, ATR-based SL/TP and trailing stops
- Applied Optuna-based Bayesian hyperparameter optimization (TPE, pruning, parallel runs) on several years of historical data to tune entry/exit parameters and risk settings
- Developed robust risk management: 40% balance position sizing, configurable leverage, ATR-driven stops, partial (50%) profit taking, and position replacement logic
- Implemented real-time monitoring loop with hourly signal evaluation, per-second position tracking, and automated CSV logging of all trades
- Added analytics and reporting: win rate, max drawdown, total/daily PnL, trade stats, and webhook-based real-time notifications
- Engineered a resilient, multi-threaded architecture with thread-safe operations, state persistence (resume from CSV), and comprehensive error handling
- Integrated with multiple exchanges and platforms (Binance, Bitget, Blofin, Bybit, MT4/MT5, others) via exchange APIs
- Performed extensive backtesting on Bitget and Blofin historical data for multiple crypto and forex pairs, including walk-forward validation to reduce overfitting

